Part III. Situational Awareness and Continuous Preparedness

This section shifts our focus from SRE culture and organizing principles to the operational realities of running reliable services. It explores how teams build deep visibility into their systems and establish and evolve practices for managing and learning from inevitable failures.

The foundation of situational awareness is understanding what "broken" actually means. In Chapter 8, "Service Level Objectives", we update the SLO framework provided in the first edition of this book. This chapter moves beyond SLO basics to describe advanced implementations, such as defining SLOs based on CUJs, handling per-customer SLOs in business-to-business (B2B) environments, and the importance of gathering client-side telemetry.

With reliability targets established, teams need the right telemetry to see if they are meeting those targets. Chapter 9, "Observability and Monitoring", details the industry shift from traditional metrics-focused monitoring to comprehensive observability. It explores how integrating time-series data, logs, traces, and continuous profiling helps manage the exponential complexity of microservices, as well as how intelligent alerting, machine learning, and dynamically generated dashboards can reduce toil and speed up time to mitigation.

When preventative measures fail and alerts fire, human responders must take control. Chapter 10, "Incident Management and On-Call", describes how to structure sustainable on-call rotations and organize effective incident response. It emphasizes that successful incident management relies not just on technical expertise but on clear coordination roles, structured triage, and effective communication and collaboration. We also touch on how AI is evolving the incident response space.

Incident response does not end when the system recovers; outages are learning opportunities. Chapter 11, "Learning from Incidents", revisits core SRE tenets of blameless postmortems and psychological safety, while discussing how these practices have matured over the last decade. It covers topics such as deeper sociotechnical system analy-

sis, customer-centric reviews, and layered assessments that ensure lessons are shared widely across the organization rather than siloed within individual teams.

Finally, Chapter 12, "Safety Engineering for SRE with STPA and CAST", challenges the software industry's traditional reliance on linear root cause analyses and chains of events. This chapter explains how to use Causal Analysis based on Systems Theory (CAST) for retrospective learning, and System-Theoretic Process Analysis (STPA) to proactively identify and eliminate design flaws before an incident occurs. By moving from reactive firefighting to proactive safety engineering, organizations can design systems that safely constrain behaviors and prevent complex failures.

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